KAIZEN MATHS

Lesson starter

Roots of Equations and Transformations of Roots

Transformed Roots

Name _____

S1

Find the new sum and product before forming the new equation.

The roots of $x^2 - x - 42 = 0$ are α and β. Form an equation whose roots are $\alpha - 3$ and $\beta - 3$.

S2

Find the new sum and product after scaling the roots.

The roots of $x^2 + 6x + 8 = 0$ are α and β. Form an equation whose roots are -2α and -2β.

S3

Use the reciprocal root relationships.

The non-zero roots of $x^2 + 11x + 30 = 0$ are α and β. Form an equation whose roots are $1\ \alpha$ and $1\ \beta$.

S4

Use the sum and product of the squared roots.

The roots of $x^2 - 12x + 32 = 0$ are α and β. Form an equation whose roots are α^2 and β^2.

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Transformed Roots

Name _____

EA1

Find the new sum and product before forming the new equation.

The roots of $x^2 - 2x - 15 = 0$ are α and β. Form an equation whose roots are $\alpha + 3$ and $\beta + 3$.

EA2

Find the new sum and product after scaling the roots.

The roots of $x^2 + 10x + 24 = 0$ are α and β. Form an equation whose roots are 3α and 3β.

Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Transformed Roots

Name _____

EB1

Use the reciprocal root relationships.

The non-zero roots of $x^2 - 9x + 14 = 0$ are α and β. Form an equation whose roots are $1\ \alpha$ and $1\ \beta$.

Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Transformed Roots

Name _____

EA1

Find the new sum and product before forming the new equation.

The roots of $x^2 - 2x - 15 = 0$ are α and β. Form an equation whose roots are $\alpha + 3$ and $\beta + 3$.

EA2

Find the new sum and product after scaling the roots.

The roots of $x^2 + 10x + 24 = 0$ are α and β. Form an equation whose roots are 3α and 3β.

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Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Transformed Roots

Name _____

EB1

Use the reciprocal root relationships.

The non-zero roots of $x^2 - 9x + 14 = 0$ are α and β. Form an equation whose roots are $1\ \alpha$ and $1\ \beta$.

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Show your working

Teacher answers & working

Roots of Equations and Transformations of Roots · Transformed Roots

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S1

Find the new sum and product before forming the new equation.

The roots of $x^2 - x - 42 = 0$ are α and β. Form an equation whose roots are α − 3 and β − 3.

Answer:

$x^2 + 5x - 36 = 0$

Working:

Let the new roots be α − 3 and β − 3.

$\text{New sum} = (\alpha - 3) + (\beta - 3) = \alpha + \beta - 6 = 1 - 6 = -5.$

New product = (α − 3)(β − 3) = αβ − 3(α + β) + 9.

$= -42 - 3(1) + 9 = -36.$

Therefore the new equation is $x^2 + 5x - 36 = 0$.

S2

Find the new sum and product after scaling the roots.

The roots of $x^2 + 6x + 8 = 0$ are α and β. Form an equation whose roots are -2α and -2β.

Answer:

$x^2 - 12x + 32 = 0$

Working:

$\text{New sum} = -2\alpha + -2\beta = -2(\alpha + \beta) = -2(-6) = 12.$

New product = (-2α)(-2β) = 4αβ = 4(8) = 32.

Use x^2 − (sum)x + product = 0.

Therefore $x^2 - 12x + 32 = 0$.

S3

Use the reciprocal root relationships.

The non-zero roots of $x^2 + 11x + 30 = 0$ are α and β. Form an equation whose roots are 1 α and 1 β .

Answer:

$30x^2 + 11x + 1 = 0$

Working:

$\text{New sum} = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-11}{30}.$

New product = $\frac{1}{\alpha\beta} = \frac{1}{30}$.

The monic equation is $x^2 - \frac{-11}{30}x + \frac{1}{30} = 0$.

Multiply every term by 30 to clear fractions.

Therefore $30x^2 + 11x + 1 = 0$.

Teacher answers & working

Roots of Equations and Transformations of Roots · Transformed Roots

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S4

Use the sum and product of the squared roots.

The roots of $x^2 - 12x + 32 = 0$ are α and β. Form an equation whose roots are α^2 and β^2.

Answer:

$x^2 - 80x + 1024 = 0$

Working:

$\text{New sum} = \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta.$

$= (12)^2 - 2(32) = 80.$

New product = $\alpha^2\beta^2 = (\alpha\beta)^2 = (32)^2 = 1024.$

Therefore the new equation is $x^2 - 80x + 1024 = 0$.

EA1

Find the new sum and product before forming the new equation.

The roots of $x^2 - 2x - 15 = 0$ are α and β. Form an equation whose roots are α + 3 and β + 3.

Answer:

$x^2 - 8x = 0$

Working:

Let the new roots be α + 3 and β + 3.

$\text{New sum} = (\alpha + 3) + (\beta + 3) = \alpha + \beta + 6 = 2 + 6 = 8.$

New product = (α + 3)(β + 3) = αβ + 3(α + β) + 9.

$= -15 + 3(2) + 9 = 0.$

Therefore the new equation is $x^2 - 8x = 0$.

EA2

Find the new sum and product after scaling the roots.

The roots of $x^2 + 10x + 24 = 0$ are α and β. Form an equation whose roots are 3α and 3β.

Answer:

$x^2 + 30x + 216 = 0$

Working:

$\text{New sum} = 3\alpha + 3\beta = 3(\alpha + \beta) = 3(-10) = -30.$

New product = (3α)(3β) = 9αβ = 9(24) = 216.

Use x^2 − (sum)x + product = 0.

Therefore $x^2 + 30x + 216 = 0$.

Teacher answers & working

Roots of Equations and Transformations of Roots · Transformed Roots

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EB1

Use the reciprocal root relationships.

The non-zero roots of $x^2 - 9x + 14 = 0$ are α and β. Form an equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

Answer:

$14x^2 - 9x + 1 = 0$

Working:

New sum $= \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{9}{14}$.

New product = $\frac{1}{\alpha\beta} = \frac{1}{14}$.

The monic equation is $x^2 - \frac{9}{14}x + \frac{1}{14} = 0$.

Multiply every term by 14 to clear fractions.

Therefore $14x^2 - 9x + 1 = 0$.